Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *AFRIDOC_MT* | ***Context:*** *Ethiopia Ministry of Health — Amharic Clinical Translation Deployment*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's output ontology operationalizes translation quality via d-BLEU, d-chrF, GPT-4o-as-judge with four explicit quality dimensions (fluency, content errors, lexical cohesion errors, grammatical cohesion errors) [Q179, Q180, Q183, Q184, Q189, Q190], and human DA on 0–100 scale [Q195, Q196]. The deployment's primary success criterion — MOH/EFDA glossary compliance — is not operationalized by any benchmark scoring function: d-BLEU/d-chrF and human DA reward similarity to the AFRIDOC-MT references and are agnostic to institutional terminology lists, and the planned terminology harmonization step (Q121) is not documented as completed. Document-level terminological consistency is partially captured by the lexical cohesion (LE) error category in GPT-4o-as-judge [Q189], but GPT-4o-as-judge is documented as unreliable for Amharic [Q99, Q218, Q219]. Faithfulness (CE) and readability (fluency) are reasonably covered. With OO marked HIGH priority by the user and glossary compliance the deployment's primary criterion, the score is 2: the benchmark's decision rules systematically misrepresent the deployment's central success criterion.

ID	Evidence
Q99	“GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages.” (p.10)
Q121	“We will provide a list of extracted terminologies soon so that you can harmonize how terminologies are translated.” (p.18)
Q179	“We compared translations performed at the sentence-level and pseudo-document level in terms of fluency, content errors, and cohesion errors—specifically lexical (LE) and grammatical (GE) errors—using the same definitions as Sun et al. (2025).” (p.22)
Q180	“GPT-4o is prompted to rate the fluency of a document on a scale from 1 to 5, where 5 indicates high fluency and 1 represents low fluency.” (p.22)
Q183	“GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model's output to the reference translation.” (p.22)
Q184	“After identifying these errors, we count all of them and compute the average across all documents, reporting that as the content error (CE).” (p.22)
Q189	“Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow.” (p.23)
Q190	“Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses.” (p.23)
Q195	“Each annotator was assigned 80 documents to evaluate, tasked with marking as many error spans as possible and rating the overall quality on a scale from 0 to 100.” (p.24)
Q196	“This annotation followed the error span annotation (ESA) (Kocmi et al., 2024) protocol as implemented within the Appraise Evaluation Framework (Federmann, 2018).” (p.24)
Q218	“These findings are consistent with the d-chrF scores for the models, but they do not align with the evaluations from GPT-4o as a judge.” (p.26)
Q219	“These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages.” (p.26)

Strengths

- Faithfulness to source meaning is well operationalized via content errors (CE) in GPT-4o-as-judge and human DA [Q183, Q184]
- Cohesion errors taxonomy (lexical and grammatical) explicitly recognizes document-level discourse phenomena [Q187, Q188, Q189, Q190]
- chrF prioritized over BLEU because it captures Amharic morphological richness — a sound choice for the target language [Q53]
- Multiple complementary scoring streams (lexical, neural, GPT-4o judge, human DA, qualitative) provide breadth [Q83, Q98]

ID	Evidence
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q83	"Outputs were evaluated using standard MT metrics, GPT-4o as a judge, and human direct assessment." (p.9)
Q98	"However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis." (p.10)
Q183	"GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model's output to the reference translation." (p.22)
Q184	"After identifying these errors, we count all of them and compute the average across all documents, reporting that as the content error (CE)." (p.22)
Q187	"Cohesion refers to how different parts of a text are connected using language structures like grammar and vocabulary." (p.23)
Q188	"It ensures that sentences flow smoothly and the text makes sense as a whole." (p.23)
Q189	"Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow." (p.23)
Q190	"Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses." (p.23)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output categories are continuous/count metrics (d-BLEU, d-chrF, fluency 1–5, CE/LE/GE counts, DA 0–100) [Q180, Q191]. Regional relevance is partial: faithfulness and fluency are universally relevant; cohesion errors capture some terminological consistency; glossary compliance is absent.

OO-2: Missing categories specific to the deployment: (a) MOH/EFDA glossary-compliance score — primary deployment success criterion is unoperationalized; (b) document-level term-consistency score that explicitly checks whether the same source term is rendered identically (LE captures vocabulary errors but not term-by-term institutional consistency); (c) readability calibrated for semi-literate patients — fluency 1–5 is not population-calibrated [DATASET CRITICAL Concern 2]; (d) Ethiopia-specific terminology coverage (endemic diseases, calendar, referral pathways).

OO-3: The cohesion-error dimensions are operationalized via GPT-4o prompts that compare model output to reference translation [Q185, Q186] — implicitly encoding the AFRIDOC-MT translators' choices as the standard, which may not align with MOH/EFDA approved terminology [DATASET-D5, D43, D47]. Fluency rated by GPT-4o without a reference [Q181] embeds whatever Amharic register GPT-4o has internalized, which may differ from Ethiopian clinical Amharic.

OO-4: Stakeholder-driven taxonomy redesign would be high value: adding (a) glossary-conformance scoring against MOH/EFDA term lists, (b) explicit term-consistency tracking, (c) MOH-translator-judged faithfulness, and (d) readability for semi-literate audiences would address the most consequential OO gaps. The Academy of Ethiopian Languages Amharic Medical Dictionary [WEB-8] and a multilingual glossary [WEB-9] exist but are not integrated.

OO-5: Structural validity is moderately compromised because the central deployment construct (institutional terminology compliance) is not represented in the scoring functions [DATASET CRITICAL Concern 2]. Content validity is also harmed by missing categories. External validity is at risk: high benchmark scores would not necessarily predict MOH-acceptable outputs.

ID	Evidence
Q121	“We will provide a list of extracted terminologies soon so that you can harmonize how terminologies are translated.” (p.18)
Q180	“GPT-4o is prompted to rate the fluency of a document on a scale from 1 to 5, where 5 indicates high fluency and 1 represents low fluency.” (p.22)
Q181	“This evaluation is conducted without providing any reference document.” (p.22)
Q183	“GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model's output to the reference translation.” (p.22)
Q185	“Cohesion: GPT-4 is prompted to rate cohesion-related mistakes, including lexical and grammatical errors, in the model's output, comparing it to the reference translation.” (p.23)
Q186	“We count each error individually, compute the average across the documents, and report them as lexical errors (LE) and grammatical errors (GE).” (p.23)
Q189	“Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow.” (p.23)
Q190	“Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses.” (p.23)
Q191	“Fluency can only have values between 1 and 5. However, the other metrics, including CE, GE, and LE, do not have a specific range and can take on any value because they are counts.” (p.24)
D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D43	ፀረ-ተህዋሲያን - ፀረ-ባክቴሪያ ፣ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርበግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)
WEB-9	Multilingual English-Amharic-Oromiffa-Somali-Afar health glossary exists on Scribd but is not integrated (https://www.scribd.com/document/416080918/English-Amharic-Oromiffa-Somali-Afar)

Information Gaps

- Whether the AFRIDOC-MT planned terminology harmonization step [Q121] was completed and what reference list was used
- Whether MOH-distributed terminology in Amharic clinical guidelines is actually formalized as a single canonical list, or whether multiple competing references exist

Requires Expert Verification

- MOH/EFDA confirmation of which Amharic glossary is the authoritative compliance reference
- Whether AFRIDOC-MT Amharic translators followed any institutional glossary during translation

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Remediation

Gap 1: No benchmark scoring function operationalizes MOH/EFDA glossary compliance — the deployment's primary success criterion is unmeasured.

Recommendation: Build a glossary-conformance scorer that extracts source clinical terms, looks them up in the Academy of Ethiopian Languages Amharic Medical Dictionary [WEB-8] and any EFDA-approved list, and scores system outputs on percentage of terms rendered with the institutionally approved Amharic equivalent. Track this metric alongside d-chrF on the Ethiopian-specific evaluation set.

Gap 2: Readability for semi-literate patients is not calibrated by the benchmark — fluency 1–5 reflects expert-translator perception, not patient comprehension.

Recommendation: Add a readability evaluation track using either an Amharic readability formula adaptation or HEW-mediated comprehension testing on a small patient sample for maternal-health booklet outputs. Report this as a separate metric alongside fluency scores.

ID	Evidence
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)